1.1 Definitions of algebra, σ-algebra

Algebra on S:

A collection Σ_0 of subsets of S is called an algebra on S

if $\begin{cases} \text{(i) } S\in\Sigma_0 \\ \text{(ii) } F\in\Sigma_0 \Rightarrow F^c := S\setminus F\in\Sigma_0 \\ \text{(iii) } F,G\in\Sigma_0 \Rightarrow F\cup G\in\Sigma_0 \end{cases}$ ⟶ finitely set operations

σ-algebra on S:

A collection Σ of subsets of S is called a σ-algebra on S if Σ is an algebra on S such that whenever $F_n\in\Sigma$ $(n\in\mathbb{N})$, then $\bigcup_n F_n\in\Sigma$ ⟶ countable collection of set operations

Measurable space

A pair (S,Σ), where S is a set and Σ is a σ-algebra on S is called a measurable space

Borel σ-algebras

$\mathcal{B} := \mathcal{B}(\mathbb{R})$

$\mathcal{B} = \sigma(\pi(\mathbb{R}))$, $\pi(\mathbb{R}) := \{(-\infty,x] : x\in\mathbb{R}\}$

Proof: $(-\infty,x] = \bigcap_{n\in\mathbb{N}}(-\infty, x+\frac{1}{n}) \in$ open sets

$\Rightarrow (-\infty,x]\in\mathcal{B}$

for any u with $u>a$, $(a,u] = (-\infty,u]\cap(-\infty,a]^c\in\sigma(\pi(\mathbb{R}))$

since, for $\varepsilon = \frac{b-a}{2}$, $(a,b) = \bigcup_n (a, b-\frac{\varepsilon}{n}]\in\sigma(\pi(\mathbb{R}))$

$\Rightarrow$ every open subset G of $\mathbb{R}$ is in $\sigma(\pi(\mathbb{R}))$

Set functions

Additive: $\mu_0 : \Sigma_0\to[0,\infty]$ is called additive if $\mu_0(\emptyset)=0$, $\forall F,G\in\Sigma_0$, $F\cap G=\emptyset \Rightarrow \mu_0(F\cup G) = \mu_0(F)+\mu_0(G)$

Countably additive

Measure Space:

Let (S,Σ) be a measurable space. Σ is a σ-algebra on S. $\mu : \Sigma\to[0,\infty]$ is called measure on (S,Σ) if μ is countably additive. The triple (S,Σ,μ) is then called a measure space.

Definitions concerning measures.

Let (S,Σ,μ) be a measure space.

μ is called $\begin{cases}\text{finite} & \text{if } \mu(S)<\infty \\ \sigma\text{-finite} & \text{if } \exists\{S_n\}, S_n\in\Sigma, \bigcup S_n = S, \mu(S_n)<\infty\ (\forall n\in\mathbb{N})\end{cases}$

Probability measure, probability triple

μ is called a probability measure if $\mu(S)=1$

then (S,Σ,μ) is called probability triple

μ-null element: an element F of Σ if $\mu(F)=0$

Moral: σ-algebras are 'difficult', but π-systems are 'easy'; so we aim to work with latter.

Lemma: Let $\Sigma = \sigma(\mathcal{I})$. Suppose that μ_1 and μ_2 are measures on (S,Σ) such that $\mu_1(S)=\mu_2(S)<\infty$ and $\mu_1=\mu_2$ on $\mathcal{I}$, then $\mu_1=\mu_2$ on Σ

Theorem: Carathéodory's Extension Theorem

Let S be a set, let Σ_0 be an algebra on S, and let $\Sigma := \sigma(\Sigma_0)$. If μ_0 is a countably additive map $\mu_0 : \Sigma_0\to[0,\infty]$, then there exists a measure μ on (S,Σ) such that $\mu=\mu_0$ on Σ_0. If $\mu_0<\infty$, this extension is unique

Lemma. Elementary inequalities

Let (S,Σ,μ) be a measure space. Then

(a) $\mu(A\cup B)\le\mu(A)+\mu(B)$ $\quad A,B\in\Sigma$

(b) $\mu(\bigcup_{i\le n}F_i)\le\sum_{i\le n}\mu(F_i)$ $\quad F_i\in\Sigma$

Furthermore, if $\mu(S)<\infty$,

(c) $\mu(A\cup B) = \mu(A)+\mu(B)-\mu(A\cap B)$

(d) (inclusion-exclusion formula): for $F_i\in\Sigma$,

$\mu(\bigcup_{i\le n}F_i) = \sum_{i\le n}\mu(F_i) - \sum_{i<j\le n}\mu(F_i\cap F_j) + \sum_{i<j<k\le n}\mu(F_i\cap F_j\cap F_k) - \cdots + (-1)^{n-1}\mu(F_1\cap\cdots\cap F_n)$

Lemma: Monotone-convergence properties of measures

(a) If $F_n\in\Sigma$ $(n\in\mathbb{N})$, $\underline{F_n\uparrow F}$, then $\mu(F_n)\uparrow\mu(F)$.
means $F_n\subseteq F_{n+1}$ $(\forall n\in\mathbb{N})$, $\bigcup F_n = F$

(b) If $G_n\in\Sigma$, $G_n\downarrow G$ and $\mu(G_k)<\infty$ for some k, then $\mu(G_n)\downarrow\mu(G)$

$(\Omega,\mathcal{F},P)$

Ω is a set called the sample space

A point ω of Ω is called a sample point

The σ-algebra $\mathcal{F}$ on Ω is called the family of events, so that an event is an element of $\mathcal{F}$, that is, an $\mathcal{F}$-measurable subset of Ω

Almost surely (a.s.)

A statement S about outcomes is said to be true almost surely (a.s.), or with probability 1 (w.p.1), if $F := \{\omega : S(\omega) \text{ is true}\}\in\mathcal{F}$ and $P(F)=1$

Proposition. If $F_n\in\mathcal{F}$ $(n\in\mathbb{N})$ and $P(F_n)=1$, $\forall n$, then $P(\bigcap_n F_n)=1$

Lim:

(a) Let $(x_n : n\in\mathbb{N})$ be a sequence of real numbers.

Def: $\limsup x_n := \inf_m\{\sup_{n\ge m}x_n\} = \downarrow\lim_m\{\sup_{n\ge m}x_n\}\in[-\infty,\infty]$.

(b) Analogously, $\liminf x_n := \sup_m\{\inf_{n\ge m}x_n\} = \uparrow\lim_m\{\inf_{n\ge m}x_n\}\in[-\infty,\infty]$

(c) x_n converges in $[-\infty,\infty]$ $\Leftrightarrow \limsup x_n = \liminf x_n$

$(E_n, \text{i.o.})$

(a) Def: $(E_n,\text{i.o.}) := (E_n \text{ infinitely often})$

$:= \limsup E_n := \bigcap_m\bigcup_{n\ge m}E_n$

$= \{\omega : \text{for every } m, \exists n(\omega)\ge m \text{ such that } \omega\in E_{n(\omega)}\}$

(b) (Reverse Fatou Lemma - needs FINITENESS of P)

$P(\limsup E_n)\ge\limsup P(E_n)$

Proof. Let $G_m := \bigcup_{n\ge m}E_n$. Then $G_m\downarrow G \Rightarrow P(G_m)\downarrow P(G)$.

But, $P(G_m)\ge\sup_{n\ge m}P(E_n)$

Hence, $P(G)\ge\downarrow\lim_m\{\sup_{n\ge m}P(E_n)\} =: \limsup P(E_n)$

First Borel-Cantelli Lemma (BC1):

Let $(E_n : n\in\mathbb{N})$ be a sequence of events such that $\sum_n P(E_n)<\infty$.

Then $P(\limsup E_n) = P(E_n,\text{i.o.}) = 0$

(E_n, ev)

(a) Def: $(E_n,\text{ev}) := (E_n \text{ eventually})$

$:= \liminf E_n := \bigcup_m\bigcap_{n\ge m}E_n$

$= \{\omega : \text{for some } m(\omega), \omega\in E_n, \forall n\ge m(\omega)\}$

$= \{\omega : \omega\in E_n \text{ for all large } n\}$

(b) $(E_n,\text{ev})^c = (E_n^c,\text{i.o.})$

(c) (Fatou's Lemma for sets - true for ALL measure spaces)

$P(\liminf E_n)\le\liminf P(E_n)$

Σ-measurable function.

Suppose that $h : S\to\mathbb{R}$. For $A\subseteq\mathbb{R}$, define $h^{-1}(A) := \{s\in S : h(s)\in A\}$. then h is called Σ-measurable if $h^{-1} : \mathcal{B}\to\Sigma$, i.e. $\forall A\in\mathcal{B}$, $h^{-1}(A)\in\Sigma$

$m\Sigma$: the class of Σ-measurable functions on S

$(m\Sigma)^+$: the class of non-negative elements on S

$b\Sigma$: the class of bounded Σ-measurable functions on S

Borel function:

A function h from a topological space S to $\mathbb{R}$ is called Borel if h is $\mathcal{B}(S)$-measurable.

Elementary Propositions on measurability:

(a) The map h^{-1} preserves all set operations:

$h^{-1}(\bigcup_\alpha A_\alpha) = \bigcup_\alpha h^{-1}(A_\alpha)$, $h^{-1}(A^c) = (h^{-1}(A))^c$

(b) If $\mathcal{C}\subseteq\mathcal{B}$ and $\sigma(\mathcal{C}) = \mathcal{B}$, then $h^{-1} : \mathcal{C}\to\Sigma \Rightarrow h\in m\Sigma$

(c) If S is topological and $h : S\to\mathbb{R}$ is continuous, then h is Borel

(d) For any measurable space (S,Σ), a function $h : S\to\mathbb{R}$ is Σ-measurable if $\{h\le c\} := \{s\in S : h(s)\le c\}\in\Sigma$ $(\forall c\in\mathbb{R})$

Lemma: $m\Sigma$ is an algebra over $\mathbb{R}$, that is, if $\lambda\in\mathbb{R}$ and $h, h_1, h_2\in m\Sigma$, then $h_1+h_2\in m\Sigma$, $h_1h_2\in m\Sigma$, $\lambda h\in m\Sigma$

Composition Lemma: If $h\in m\Sigma$, $f\in m\mathcal{B}$, then $f\circ h\in m\Sigma$

Lemma on measurability of infs, lim infs of functions

Let $(h_n : n\in\mathbb{N})$ be a sequence of elements of $m\Sigma$.

Then (i) $\inf h_n$, (ii) $\liminf h_n$, (iii) $\limsup h_n$ are Σ-measurable.

Further, (iv) $\{s : \lim h_n(s) \text{ exists in } \mathbb{R}\}\in\Sigma$

Random variable:

Let $(\Omega,\mathcal{F})$ be our (sample space, family of events). A random variable is an element of $m\mathcal{F}$.

Thus, $X : \Omega\to\mathbb{R}$, $X^{-1} : \mathcal{B}\to\mathcal{F}$

Def. if we have a collection $(Y_\gamma : \gamma\in C)$ of maps $Y_\gamma : \Omega\to\mathbb{R}$, then $\mathcal{Y} := \sigma(Y_\gamma : \gamma\in C)$ is defined to be the smallest σ-algebra $\mathcal{Y}$ on Ω such that each map Y_γ is $\mathcal{Y}$-measurable

If X is a r.v. then $\sigma(X)\subseteq\mathcal{F}$

Def. Suppose that X is a r.v. carried by some probability triple $(\Omega,\mathcal{F},P)$

We have $\Omega\xrightarrow{X}\mathbb{R}$, $[0,1]\xleftarrow{P}\mathcal{F}\xleftarrow{X^{-1}}\mathcal{B}$, or indeed $[0,1]\xleftarrow{P}\sigma(X)\xleftarrow{X^{-1}}\mathcal{B}$

Define the law $\mathcal{L}_X$ of X by: $\mathcal{L}_X := P\circ X^{-1}$, $\mathcal{L}_X : \mathcal{B}\to[0,1]$

Since $\mathcal{L}_X$ is a probability measure on $(\mathbb{R},\mathcal{B})$, $\pi(\mathbb{R})$ is a π-system which generates $\mathcal{B}$, based on Uniqueness Lemma, $\mathcal{L}_X$ is determined by $F_X : \mathbb{R}\to[0,1]$ defined as $F_X(c) := \mathcal{L}_X(-\infty,c] = P(X\le c)$

F_X is called the distribution function of X

Properties of distribution function.

(a) $F : \mathbb{R}\to[0,1]$, $F\uparrow$ (that is, $x\le y\Rightarrow F(x)\le F(y)$)

(b) $\lim_{x\to\infty}F(x)=1$, $\lim_{x\to-\infty}F(x)=0$

(c) F is right-continuous

Skorokhod representation of a random variable with pre-scribed DF

Let $F : \mathbb{R}\to[0,1]$ is a distribution function.

We can construct a r.v. with DF F carried by $(\Omega,\mathcal{F},P) = ([0,1],\mathcal{B}[0,1],\text{Leb})$

Def: $X^+(\omega) := \inf\{z : F(z)>\omega\} = \sup\{y : F(y)\le\omega\}$

$X^-(\omega) := \inf\{z : F(z)\ge\omega\} = \sup\{y : F(y)<\omega\}$

then, $\omega\le F(c) \Rightarrow X^-(\omega)\le c$

$z>X^-(\omega) \Rightarrow F(z)\ge\omega$

$\Rightarrow F(X^-(\omega))\ge\omega$ and $(X^-(\omega)\le c)\Rightarrow(\omega\le F(X^-(\omega))\le F(c))$

thus, $P(X^-\le c) = F(c)$ i.e. variable X^- has DF F

Similarly, X^+ also has DF F.

and $P(X^+ = X^-) = 1$

Independent σ-algebras:

Sub-σ-algebras $\mathcal{G}_1, \mathcal{G}_2, \dots$ of $\mathcal{F}$ are called independent if, whenever $G_i\in\mathcal{G}_i$ $(i\in\mathbb{N})$ and $i_1,\dots,i_n$ are distinct, then $P(G_{i_1}\cap\cdots\cap G_{i_n}) = \prod_{k=1}^n P(G_{i_k})$

Independent r.v.:

Random variables $X_1, X_2, \dots$ are called independent if the σ-algebras $\sigma(X_1), \dots$ are independent.

Independent events

Events $E_1, E_2, \dots$ are called independent if the σ-algebras $\mathcal{E}_1, \mathcal{E}_2, \dots$ are independent, where $\mathcal{E}_n = \sigma(I_{E_n}) = \sigma(\{\emptyset, E_n, \Omega\setminus E_n, \Omega\})$

Lemma. Suppose that $\mathcal{G}$ and $\mathcal{H}$ are sub-σ-algebras of $\mathcal{F}$, and that $\mathcal{I}$ and $\mathcal{J}$ are π-systems with $\sigma(\mathcal{I}) = \mathcal{G}$, $\sigma(\mathcal{J}) = \mathcal{H}$. Then $\mathcal{G}$ and $\mathcal{H}$ are independent if and only if $\mathcal{I}$ and $\mathcal{J}$ are independent in that $P(I\cap J) = P(I)P(J)$, $I\in\mathcal{I}$, $J\in\mathcal{J}$

Second Borel-Cantelli Lemma (BC2)

If $(E_n : n\in\mathbb{N})$ is a sequence of independent events, then

$\sum P(E_n) = \infty \Rightarrow P(E_n,\text{i.o.}) = P(\limsup E_n) = 1$

Stochastic processes

A stochastic process Y parametrized by a set C is a collection $Y = (Y_\gamma : \gamma\in C)$ of RVs on same triple $(\Omega,\mathcal{F},P)$

We think of X_n as the value of the process X at time n

For $\omega\in\Omega$, the map $n\mapsto X_n(\omega)$ is called the sample path of X corresponding to the sample point ω.

Markov chain:

Let E be a finite or countable set. Let $P = (p_{ij} : i,j\in E)$ be a stochastic $E\times E$ matrix, so that for $i,j\in E$, we have $\begin{cases}p_{ij}\ge 0 \\ \sum_k p_{ik} = 1\end{cases}$

Let μ be a probability measure on E, so that μ is specified by the values $\mu_i := \mu(\{i\})$ $(i\in E)$

By a time-homogeneous Markov chain $Z = (Z_n : n\in\mathbb{Z}^+)$ on E with initial distribution μ and 1-step transition matrix P is meant a stochastic process Z such that, whenever $n\in\mathbb{Z}^+$, $i_0, i_1, \dots, i_n\in E$, $P(Z_0=i_0; Z_1=i_1; \dots; Z_n=i_n) = \mu_{i_0}p_{i_0i_1}\cdots p_{i_{n-1}i_n}$

Tail σ-algebras:

Let $X_1, X_2, \dots$ be random variables. Def

$\mathcal{T}_n := \sigma(X_{n+1}, X_{n+2}, \dots)$, $\mathcal{T} := \bigcap_n\mathcal{T}_n$

The σ-algebra $\mathcal{T}$ is called the tail σ-algebra of the sequence $(X_n : n\in\mathbb{N})$

Kolmogorov's 0-1 Law:

Let $(X_n : n\in\mathbb{N})$ be a sequence of independent random variables, and let $\mathcal{T}$ be the tail σ-algebra of $(X_n : n\in\mathbb{N})$. Then $\mathcal{T}$ is P-trivial: that is, (i) $F\in\mathcal{T}\Rightarrow P(F)=0$ or $P(F)=1$,

(ii) if ξ is a $\mathcal{T}$-measurable r.v., then, ξ is almost deterministic in that for some constant c in $[-\infty,\infty]$, $P(\xi=c)=1$

Monotone-Convergence Theorem (MON)

If (f_n) is a sequence of elements of $(m\Sigma)^+$ such that $f_n\uparrow f$,

then $\mu(f_n)\uparrow\mu(f)\le\infty$, or, in other notation, $\int_S f_n(s)\mu(ds)\uparrow\int_S f(s)\mu(ds)$

Dominated-Convergence Theorem (DOM)

Suppose that $f_n, f\in m\Sigma$, that $f_n(s)\to f(s)$ for every s in S and that the sequence (f_n) is dominated by an element g of $\mathcal{L}^1(S,\Sigma,\mu)^+$:

$|f_n(s)|\le g(s)$, $\forall s\in S$, $\forall n\in\mathbb{N}$ where $\mu(g)<\infty$.

then $f_n\to f$ in $\mathcal{L}^1(S,\Sigma,\mu)$: that is, $\mu(|f_n-f|)\to 0$, whence $\mu(f_n)\to\mu(f)$

Scheffé's Lemma (SCHEFFÉ)

Suppose that $f_n, f\in\mathcal{L}^1(S,\Sigma,\mu)^+$, in particular, f_n and f are non-negative.

Suppose that $f_n\to f$ (a.e.). Then $\mu(|f_n-f|)\to 0$ i.o.i $\mu(|f_n|)\to\mu(|f|)$

Def. For a RV $X\in\mathcal{L}^1 = \mathcal{L}^1(\Omega,\mathcal{F},P)$, we define the expectation $E(X)$ of X by

$E(X) := \int_\Omega X\,dP = \int_\Omega X(\omega)P(d\omega)$

Convergence Theorems (Expectation):

MON: if $0\le X_n\uparrow X$, then $E(X_n)\uparrow E(X)\le\infty$

FATOU: if $X_n\ge 0$, then $E(X)\le\liminf E(X_n)$

DOM: if $|X_n(\omega)|\le Y(\omega)$ $\forall(n,\omega)$, where $E(Y)<\infty$, then $E(|X_n-X|)\to 0$ so that $E(X_n)\to E(X)$

SCHEFFÉ: if $E(|X_n|)\to E(|X|)$, then $E(|X_n-X|)\to 0$

BDD: if for some finite constant K, $|X_n(\omega)|\le K$, $\forall(n,\omega)$, then $E(|X_n-X|)\to 0$

The notation $E(X;F)$

we define $E(X;F) := \int_F X(\omega)P(d\omega) =: E(XI_F)$

Markov's Inequality:

Support that $Z\in m\mathcal{F}$ and that $g : \mathbb{R}\to[0,\infty)$ is $\mathcal{B}$-measurable and non-decreasing, then

$Eg(Z)\ge E(g(Z); Z\ge c)\ge g(c)P(Z\ge c)$

Jensen's inequality for convex functions

Suppose that $c : G\to\mathbb{R}$ is a convex function on an open subinterval G of $\mathbb{R}$ and that X is a r.v. such that $E(|X|)<\infty$, $P(X\in G)=1$, $E|c(X)|<\infty$

then, $Ec(X)\ge c(E(X))$

Monotonicity of $\mathcal{L}^p$ norms

For $1\le p\le\infty$, we say that $X\in\mathcal{L}^p = \mathcal{L}^p(\Omega,\mathcal{F},P)$ if $E(|X|^p)<\infty$ and we define $\|X\|_p := \{E(|X|^p)\}^{\frac{1}{p}}$

The monotonicity property:

if $1\le p\le r<\infty$ and $Y\in\mathcal{L}^r$, then $Y\in\mathcal{L}^p$ and $\|Y\|_p\le\|Y\|_r$

Vector-space property of $\mathcal{L}^p$

for $a,b\in\mathbb{R}^+$, we have $(a+b)^p\le[2\max(a,b)]^p\le 2^p(a^p+b^p)$

so $\mathcal{L}^p$ is a vector space

The Schwarz Inequality:

if X and Y are in $\mathcal{L}^2$, then $XY\in\mathcal{L}^1$,

and $|E(XY)|\le E(|XY|)\le\|X\|_2\|Y\|_2$

Some Inequalities:

Let (S,Σ,μ) be a measure space. Suppose that $p>1$ and $p^{-1}+q^{-1}=1$.

if $f,g\in\mathcal{L}^p(S,\Sigma,\mu)$, $h\in\mathcal{L}^q(S,\Sigma,\mu)$, then

(a) Hölder's inequality: $fh\in\mathcal{L}^1(S,\Sigma,\mu)$ and

$|\mu(fh)|\le\mu(|fh|)\le\|f\|_p\|h\|_q$

(b) Minkowski's inequality: $\|f+g\|_p\le\|f\|_p+\|g\|_p$

Completeness of $\mathcal{L}^p$ $(1\le p<\infty)$

分布函数的弱收敛：设$\{F_n\}$为分布函数序列，若存在一个右连续且递增的函数F，使在F的任一连续点x处有$\lim_{n\to\infty}F_n(x)=F(x)$，则称$F_n$弱收敛于$F$，记为$F_n\xrightarrow{w}F$

依分布收敛：设$\{X_n\}$分布函数为F_n，X分布函数为F，若$F_n\xrightarrow{w}F$，则称X_n依分布收敛于X，记为$X_n\xrightarrow{D}X$

记$\mathrm{Prob}(\mathbb{R})$为$\mathbb{R}$上的概率测度全体，$C_b(\mathbb{R})$为$\mathbb{R}$上有界连续函数全体

概率测度弱收敛：设$\{\mu_n\}_{n\ge1}$是$\mathrm{Prob}(\mathbb{R})$中的一个序列，$\mu\in\mathrm{Prob}(\mathbb{R})$，称$\mu_n$弱收敛至$\mu$，如果$\mu_n(f)\to\mu(f)$，$\forall f\in C_b(\mathbb{R})$，记为$\mu_n\xrightarrow{w}\mu$

Helly-Bray Lemma:

假设$\{F_n\}$，F是分布函数，$\{\mu_n\}$，μ是对应概率测度，则$F_n\xrightarrow{w}F$当且仅当

$\mu_n\xrightarrow{w}\mu$，即$\mu_n(f)=\int_{\mathbb{R}}f(x)dF_n(x)\longrightarrow\int_{\mathbb{R}}f(x)dF(x)=\mu(f)$，$\forall f\in C_b(\mathbb{R})$

推论：假设$\{F_n\}$，F是分布函数，若对$\forall$子列$\{F_{n(m)}\}$均存在子列的子列$\{F_{n(m_k)}\}$使得$F_{n(m_k)}\xrightarrow{w}F$，则$F_n\xrightarrow{w}F$

Ext Theorem: if $y_n\in\mathbb{R}$，若$\forall$子列$y_{n(m)}$均存在子列的子列$y_{n(m_k)}$使得$y_{n(m_k)}\to y$，则$y_n\to y$

Skorokhod representation

Theorem: Suppose that $\{F_n\}$ is a sequence of DFs on $\mathbb{R}$, that F is a DF on $\mathbb{R}$ and that $F_n(x)\to F(x)$ at every point x of continuity of F.

Then there exists a probability triple $(\Omega,\mathcal{F},P)$ carrying a sequence (X_n) of RVs and also a RV X such that $F_n = F_{X_n}$, $F = F_X$, and $X_n\to X$ a.s.

Tightness

A sequence $\{F_n\}$ of DFs is called tight if, given $\varepsilon>0$, there exists $K(\varepsilon)>0$, such that $\mu_n[-K,K] = F_n(K)-F_n(-K)>1-\varepsilon$

Lemma:

$\{F_n\}$是胎紧的当且仅当$\{F_n\}$的任一弱收敛子列，其弱收敛极限为分布函数

连续性定理：设 DF：$\{F_n\}$和F，对应特征函数为$\{\phi_n\}$，ϕ

(a) 若$F_n\xrightarrow{w}F$，则$\phi_n\to\phi$

(b) 若$\phi_n\to\varphi$且φ在零点连续，则φ是某个分布函数F的特征函数

Def (Kolmogorov, 1933) Theorem:

Let $(\Omega,\mathcal{F},P)$ be a triple, and X a RV with $E(|X|)<\infty$. Let $\mathcal{G}$ be a sub-σ-algebra of $\mathcal{F}$. Then there exist a RV Y such that (a) $Y\in m\mathcal{G}$ (b) $E(|Y|)<\infty$ (c) for every set G in $\mathcal{G}$, we have $\int_G Y\,dP = \int_G X\,dP$, $\forall G\in\mathcal{G}$

Moreover, if $\tilde{Y}$ is another RV with these properties, then $P(\tilde{Y}=Y)=1$

A RV Y with properties (a)-(c) is called a version of the conditional expectation $E(X|\mathcal{G})$ of X given $\mathcal{G}$, and we write $Y = E(X|\mathcal{G})$, a.s.

Properties of Conditional expectation

$\mathcal{G}$ and $\mathcal{H}$ are sub-σ-algebras of $\mathcal{F}$. $E(|X|)<\infty$

(a) If Y is any version of $E(X|\mathcal{G})$ then $E(Y)=E(X)$

(b) If $X\in m\mathcal{G}$, then $E(X|\mathcal{G}) = X$, a.s.

Proof. $E(E(X|\mathcal{G});G) = E(X;G)$

(c) (Linearity) $E(aX_1+bX_2|\mathcal{G}) = aE(X_1|\mathcal{G})+bE(X_2|\mathcal{G})$, a.s.

(d) (Positivity) If $X\ge 0$, then $E(X|\mathcal{G})\ge 0$, a.s.

(e) (cMON) If $0\le X_n\uparrow X$, then $E(X_n|\mathcal{G})\uparrow E(X|\mathcal{G})$, a.s.

(f) (cFATOU) If $X_n\ge 0$, then $E[\liminf X_n|\mathcal{G}]\le\liminf E(X_n|\mathcal{G})$, a.s.

(g) (cDOM) If $|X_n(\omega)|\le V(\omega)$, $\forall n$, $EV<\infty$, and $X_n\to X$, a.s., then $E(X_n|\mathcal{G})\to E(X|\mathcal{G})$, a.s.

(h) (cJensen) If $c : \mathbb{R}\to\mathbb{R}$ is convex, and $E(|c(X)|)<\infty$, then $E(c(X)|\mathcal{G})\ge c(E(X|\mathcal{G}))$, a.s.

Important corollary: $\|E(X|\mathcal{G})\|_p\le\|X\|_p$ for $p\ge 1$

(i) (Tower Property) If $\mathcal{H}$ is a sub-σ-algebra of $\mathcal{G}$, then $E[E(X|\mathcal{G})|\mathcal{H}] = E(X|\mathcal{H})$, a.s.

(j) If Z is $\mathcal{G}$-measurable and bounded, then $E(ZX|\mathcal{G}) = ZE(X|\mathcal{G})$, a.s.

If $p>1$, $p^{-1}+q^{-1}=1$, $X\in\mathcal{L}^p(\Omega,\mathcal{F},P)$ and $Z\in\mathcal{L}^q(\Omega,\mathcal{G},P)$, then it again holds

If $X\in(m\mathcal{F})^+$, $Z\in(m\mathcal{G})^+$, $E(X)<\infty$ and $E(ZX)<\infty$, then it holds

(k) (Role of Independence) If $\mathcal{H}$ is independent of $\sigma(\sigma(X),\mathcal{G})$, then $E[X|\sigma(\mathcal{G},\mathcal{H})] = E(X|\mathcal{G})$, a.s.

Filtered space $(\Omega,\mathcal{F},\{\mathcal{F}_n\},P)$:

$\{\mathcal{F}_n : n\ge 0\}$ is a filtration, that is, an increasing family of sub-σ-algebras of $\mathcal{F}$: $\mathcal{F}_0\subseteq\mathcal{F}_1\subseteq\cdots\subseteq\mathcal{F}$.

Def. $\mathcal{F}_\infty := \sigma(\bigcup_n\mathcal{F}_n)\subseteq\mathcal{F}$

Adapted process

A process $X = (X_n : n\ge 0)$ is called adapted (to the filtration $\{\mathcal{F}_n\}$) if for each n, $X_n\in m\mathcal{F}_n$

Martingale.

A process X is called a martingale (relative to $(\{\mathcal{F}_n\},P)$) if (i) X is adapted (ii) $\forall n$, $E(|X_n|)<\infty$ (iii) $E[X_n|\mathcal{F}_{n-1}] = X_{n-1}$, a.s. $(n\ge 1)$

supermartingale ($E[X_n|\mathcal{F}_{n-1}]\le X_{n-1}$, a.s. $(n\ge 1)$)

submartingale ($E[X_n|\mathcal{F}_{n-1}]\ge X_{n-1}$, a.s. $(n\ge 1)$)

$\Rightarrow$ X is a supermartingale iff. $-X$ is a submartingale

X is a martingale iff. X is both a supermartingale and a submartingale

We call a process $C = (C_n : n\in\mathbb{N})$ previsible if C_n is $\mathcal{F}_{n-1}$ measurable $(n\ge 1)$

Stopping time

A map $T : \Omega\to\{0,1,2,\dots;\infty\}$ is called a stopping time if

(a) $\{T\le n\} = \{\omega : T(\omega)\le n\}\in\mathcal{F}_n$, $\forall n\le\infty$

or (b) $\{T=n\} = \{\omega : T(\omega)=n\}\in\mathcal{F}_n$, $\forall n\le\infty$

Doob's Optional-Stopping Theorem:

(a) Let T be a stopping time, X be a martingale. Then X_T is integrable and $E(X_T)\le E(X_0)$ in each of the following situations:

(i) T is bounded (ii) X is bounded (iii) $E(T)<\infty$ and $\exists K\in\mathbb{R}^+$, $|X_n(\omega)-X_{n-1}(\omega)|\le K$ $\forall(n,\omega)$

(b) If any of the conditions (i)-(iii) holds and X is a martingale, then $E(X_T) = E(X_0)$